Step Smart:

Injury Preventive Insole

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Other Members: Evan Bohler and Charlie Whittleman

Overall Problem

Athletes experience joint pain and injury from improper foot placement

- Many athletes suffer injuries due to improper foot posture, leading to inflammation, pain, and setbacks.

- Some injuries may force athletes to pause or even end their training or careers.

- According to the American Academy of Orthopedic Surgeons, 25% of sports injuries are foot and ankle related.

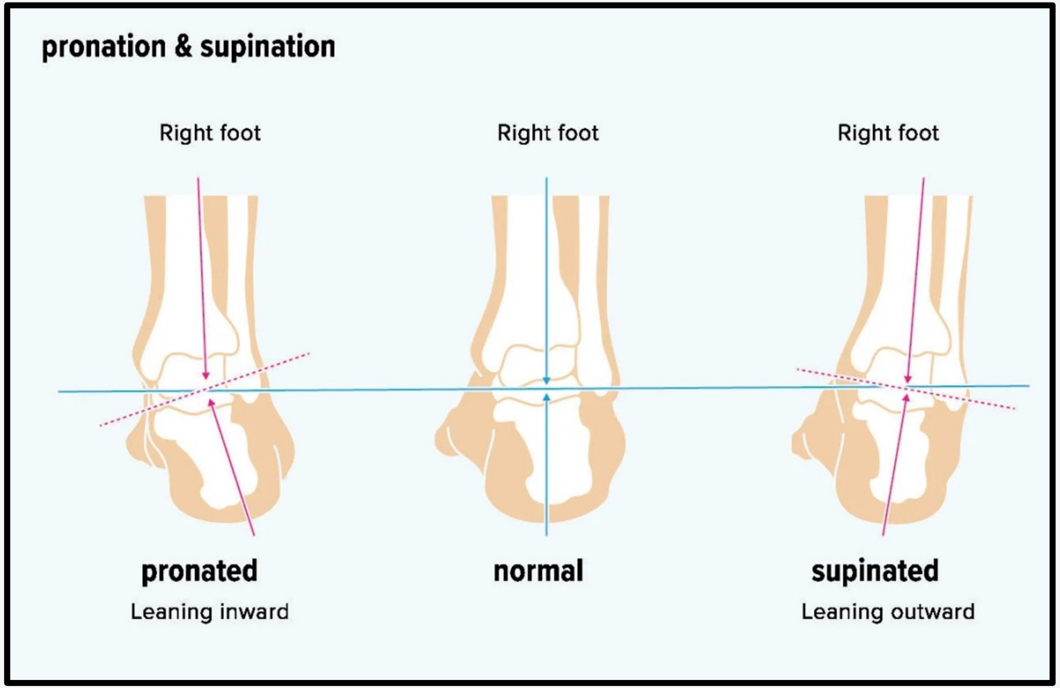
- We aim to design a solution that corrects poor placement and prevents injuries before they happen.

Foot Posture

- Overpronation occurs when the foot rolls inward excessively, while oversupination occurs when the foot rolls outward

- Places unnecessary stress on the body, leading to inflammation, pain, and potentially permanent injuries

- What if we could track weight distribution and foot posture in real-time?

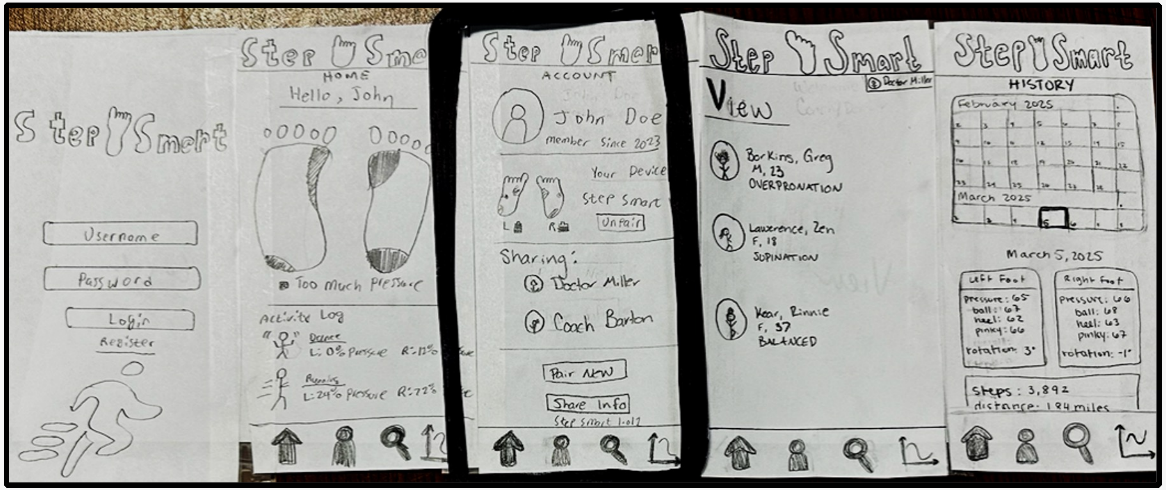


https://www.healthline.com/health/bone-health/whats-the-difference-between-supination-and-pronation#the-foot

Initial Paper Prototype

App Prototype:

- Multiple screens such as:
    - Log In
    - Home screen to view real time position
    - Account screen to view device list and sharing list
    - View screen (for coaches) to view team members wearing the device
    - History screen to track foot positioning over time



Insert Prototype:

- Different sizes to account for different shoe types such as ballet slippers vs soccer cleats.
- Sensors on main parts of the foot to read pressure placement to calculate over pronation vs supination.
- Made thin and flexible to be inconspicuous once in the shoe

Know How One’s Feet Are Positioned During Activities

The user must:

- Put the insoles in their shoes during the activity they wish to track
- During/after the activity, either the user or their coach can open their Step Smart app
- Posture data is synced in real time
- User is able to easily read the graphs and reposition their feet accordingly

Use an App to See Health and Fitness Data Visuals

Within our intuitive app:

- First, the user must log into the app
- Account page
- Home tab
    - Shows a clear visualization map of the foot
    - Keeps an activity log
- History tab
    - Detailed information on their posture trends
    - View data from previous days using the calendar
- View Page
    - Coach can view data of team members

Testing & Results

Methods

We conducted three **usability tests**.

To ensure our app was intuitive we asked three people to use our prototype within a variety of circumstances. Then we adjusted our final design.

Participants

Participant 1:
18 year old dancer of 15 years and volleyball player of 5

Participant 2:
A computer scientist classmate in his 20s, he regularly works out

Participant 3:
A computer scientist classmate in his 20s with a history of knee problems

Key Findings

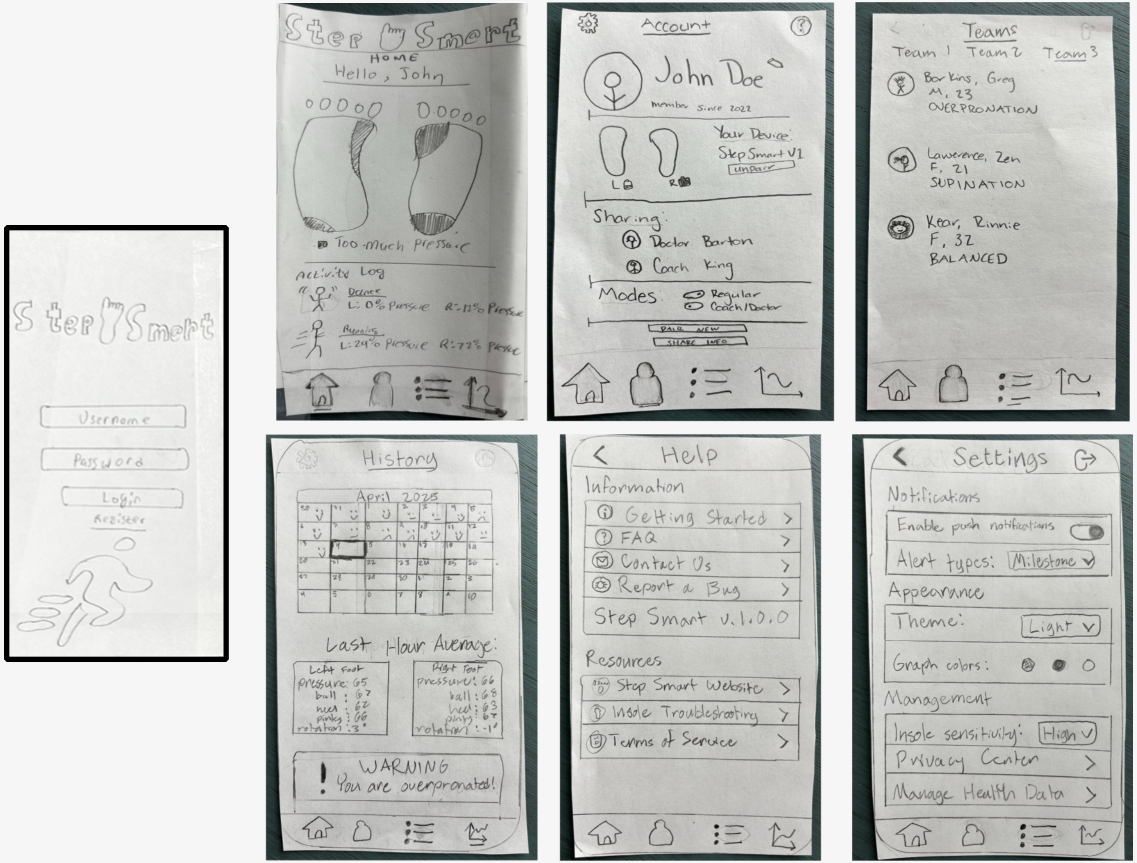
To make use of the app as seamless as possible, we had to alter our **view** tab as well as make **navigation** between screens simple and instinctive as possible

Final Paper Prototype

App Prototype:

Multiple screens such as:

- Log In
- Home screen to view real time position
- Account screen to view device list, sharing list, and **toggle coach mode**
- **Teams screen** (for coach mode) to view team members wearing device
- History screen to track foot positioning over time
- Push notifications
- **Help & settings menu**
- **Improved icons, app bar, visuals, and intuitiveness**



Insert Prototype:

- Different sizes to account for different shoe types such as ballet slippers vs soccer cleats.
- Sensors on main parts of the foot to read pressure placement to calculate over pronation vs supination.
- Made thin and flexible to be inconspicuous once in the shoe
- **Added grip to the underside to prevent movement in the shoe for maximum comfort**

Know How One’s Feet Are Positioned During Activities

- User must simply wear the padded insert during an activity and the app will show where the most pressure resides on their feet in real-time.
- Knowing where too much or too little pressure allows the user to reposition their feet to ensure injuries are unlikely.
- If the user is consistently wearing the insert, they will be able to easily tell which activities are causing them pain through our user-friendly graphs and push notifications.

Use an App to See Health and Fitness Data Visuals

- **Improved data presentation**
  - **Added feedback on the history calendar for good/bad days**
  - **Live notifications for when foot position is improper**
- View page changed to Teams
  - Allows coaches to switch between different teams and view athlete data more intuitively
- **Improved user experience**
  - Improved top bar: page titles, sign-in/sign-out options, back button, links to settings and help pages
  - **Separate user/coach app experiences**

Digital Mockup

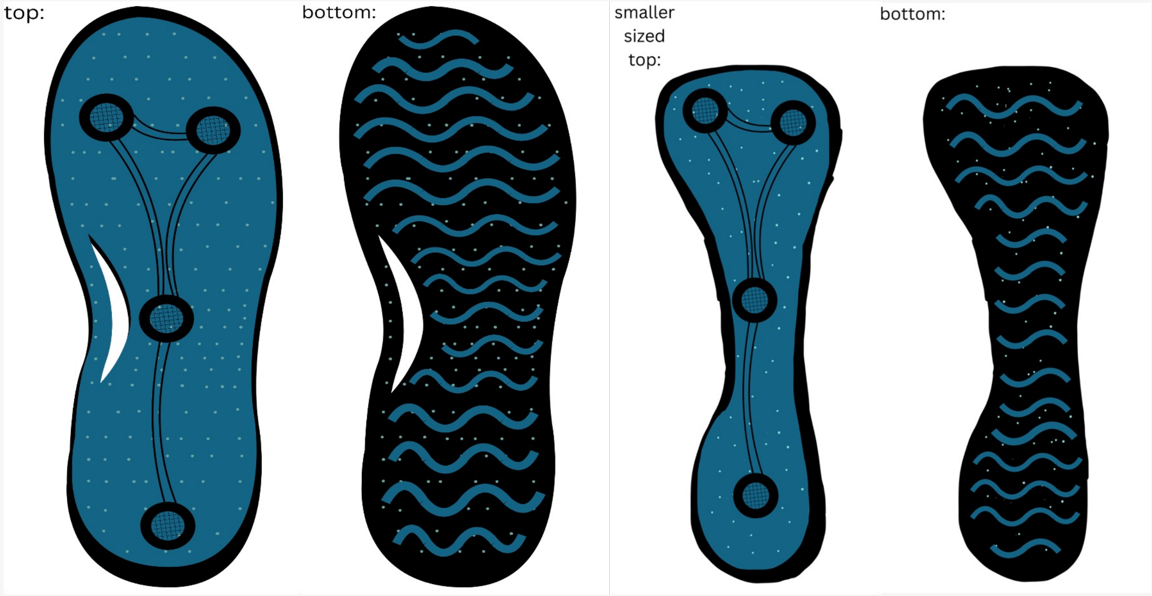
App Prototype:

Same screens as before:

- **Improved tab icons**
- **Improved top bar buttons: added notifications and put the settings/help button in the Account menu**
- **Made it possible to add new members/teams in the Teams menu**
- **Added sign up and forgot password buttons to login**
- **Overall, improved visual aspects and intuitiveness**

Insert Prototype:

- Different sizes to account for different shoe types such as ballet slippers vs soccer cleats.
- Sensors on main parts of the foot to read pressure placement to calculate over pronation vs supination.
- Made thin, flexible, and with button grip to be inconspicuous and secure once in the shoe
- **Added an extra sensor on the arch of the foot for maximum pressure assessment**



Know How One’s Feet Are Positioned During Activities

- **Fourth sensor added**
  - More precise pressure tracking
- **Notification button** added to top navigation bar
  - Grants users **quick access to notification** history at any time
  - If dangerous pressure levels are detected, **users can prevent injuries before they occur.**

Use an App to See Health and Fitness Data Visuals

- User Experience Improvements
  - **Cleaner interface** with consistent styling and color scheme
    - **Overall improves user experience in viewing data**
  - Underline indicator to highlight the current tab
  - Settings and help buttons together on account page
  - Replaced the history page icon with a calendar icon
  - Added “+” button in the teams page to create a team and/or add members
  - Added registration and forgot password button to login screen

Summary

Overall Lesson

Creating something from scratch takes lots of revisions and learning through others

- First design is never going to be the final product

- Multiple series of interviews and testing need to be done to get a well rounded opinion of product usability
    - Multiple people with different backgrounds need to be assessed to see the picture from all sides

- Listen to all critiques and advice with open ears and try to walk in the user's shoes to find what works best for the product

Thank you